A_{TM} ∾ $HALT_{TM}$, E_{TM}, $REGULAR_{TM}$, EQ_{TM}

Try to show that if EQ_{TM} is decidable, E_{TM} is decidable.

Assume that EQ_{TM} is decidable by TM R.

Construct TM S to decide $E_{TM} = \{\langle M \rangle \mid M \text{ is a TM and } L(M) = \emptyset\}$

S = "On input $\langle M \rangle$ where M is a TM:

1. Construct TM M' as follows:

   M' = " On input x:
   1. reject "

   } $L(M') = \emptyset$

2. Run R on input $\langle M, M' \rangle$
3. If R accepts $\langle M, M' \rangle$, accept
   If R rejects $\langle M, M' \rangle$, reject "

1. If $\langle M \rangle \in E_{TM}$, S must accept $\langle M \rangle$
2. If $\langle M \rangle \notin E_{TM}$, S must reject $\langle M \rangle$.

$ONE_{TM} = \{\langle M \rangle \mid M \text{ is a TM and } |L(M)| = 1\}$

$B = \{\langle M_1, M_2\rangle \mid M_1 \text{ and } M_2 \text{ are TMs and } L(M_1) \cup L(M_2) = \Sigma^*\}$ Show that B is undecidable by showing that if B is decidable, A_{TM} is decidable.

Assume that B is decidable. Since B is decidable, there exists a TM M_B that decides B.

Construct TM S to decide A_{TM}

S = "On input $\langle M, w\rangle$ where M is a TM and w is a string:

1. Construct TM M_1 and M_2 as follows:

   M_1 = "On input x:
   1. reject " } $L(M_1) = \emptyset$

   M_2 = "On input x:
   1. Run M on input w
   2. If M accepts w, accept
      If M rejects w, reject "

2. Run M_B on input $\langle M_1, M_2\rangle$
3. If M_B accepts $\langle M_1, M_2\rangle$, accept
   If M_B rejects $\langle M_1, M_2\rangle$, reject "

Since M_B accepts $\langle M_1, M_2\rangle$, S accepts $\langle M, w\rangle$
(M_B rejects, S rejects)

2.
1. If $\langle M, w\rangle \notin A_{TM}$, S will reject $\langle M, w\rangle$.

Assume that $\langle M, w\rangle \notin A_{TM}$. Since $\langle M, w\rangle \in A_{TM}$, M accept (does not accept) w. Since M accepts w, $L(M_2) = \Sigma^*$ ($\emptyset^*$). Since $L(M_1) = \emptyset$, $L(M_1) \cup L(M_2) = \emptyset \neq \Sigma^*$. Since $L(M_1) \cup L(M_2) = \Sigma^*$, $\langle M_1, M_2\rangle \notin B$. Since M_B is a decider for B, by running M_B on input $\langle M_1, M_2\rangle$, M_B will reject $\langle M_1, M_2\rangle$ (accept ↑)

Mapping Reducibility

$A \leq_m B$ if $f: \Sigma^* \to \Sigma^*$ exists and satisfies
↳ a reduction from A to B

$$w \in A \leftrightarrow \underbrace{f(w)}_{F(w)} \in B \quad *$$

If $A \leq_m B$ and B is decidable, A is decidable
(P) (∧) (Q) (→) (R)

$(P \wedge Q) \to R \equiv (P \wedge \neg R) \to \neg Q$

The reduction f exists TM M_B that decides B
↑
TM F

1. If $w \in A$, M_A must accept w
2. If $w \notin A$, M_A must reject w

Construct M_A to decide A

M_A = "On input w:
1. Run F on input w to obtain F(w)
2. Run M_B on input F(w)
3. If M_B accepts F(w), accept
   If M_B rejects F(w), reject"

If $A \leq_m B$ and A is undecidable, B is undecidable.

If $A_{TM} \leq_m B$ and A_{TM} is undecidable, B is undecidable

fact exist

$$\underbrace{\langle M, w \rangle}_{w} \in A_{TM} \leftrightarrow F(\underbrace{\langle M, w \rangle}_{w}) \in B$$

Prove that EQ_{TM} is undecidable using mapping reducibility

If $A_{TM} \leq_m EQ_{TM}$ and A_{TM} is undecidable, EQ_{TM} is undecidable

$EQ_{TM} = \{\langle M_1, M_2 \rangle \mid \ldots\}$

$\langle M, w \rangle \in A_{TM} \leftrightarrow F(\langle M, w \rangle) \in EQ_{TM}$

$F(\langle M, w \rangle) = \langle M_1, M_2 \rangle$

F = "On input $\langle M, w \rangle$ where M is a TM and w is a string:

1. Construct TMs M_1 and M_2 as follows:

   M_1 = "On input x:
   1. accept"

   M_2 = "On input x:
   1. Run M on input w
   2. If M accepts w, accept
      If M rejects w, reject"

2. Output $\langle M_1, M_2 \rangle$

1. If $\langle M, w \rangle \in A_{TM}$, $F(\langle M, w \rangle) \in EQ_{TM}$
2. If $\langle M, w \rangle \notin A_{TM}$, $F(\langle M, w \rangle) \notin EQ_{TM}$

If $A_{TM} \leq_m HALT_{TM}$ and A_{TM} is undecidable, $HALT_{TM}$ is undecidable.

* $\langle M, w \rangle \in A_{TM} \leftrightarrow F(\langle M, w \rangle) \in HALT_{TM}$ $= \{\langle M, w \rangle | M \text{ is a TM that halts on } w\}$

F = "On input $\langle M, w \rangle$ where M is a TM and w is a string:

1. Construct TM M' as follows:

   M' = "On input x:
   1. Run M on input w
   2. If M accepts w, accept
      If M rejects w, enter infinite loop"

2. Output $\langle M', \varepsilon \rangle$"

$f: \Sigma^* \to \Sigma^*$

$w \in A \leftrightarrow f(w) \in B$